

“WE MAKE DEVELOPMENT KITS AVAILABLE TO DESIGNERS AT A LOW PRICE OF \$30”



Be it a manufacturer or a design house in the electronics space, everyone wants to know about innovations that will be game changers in the semiconductor space and whether India is ready to become the ace player beating out others. Rene Penning de Vries, senior vice-president and chief technology officer of NXP Semiconductors responsible for overseeing the product creation process focusing on the areas of innovations, research and technology, talks to EFY's Abhijit Paul Choudhury and Vanisha Joseph about trends to be watched for in the semiconductor industry and how India can be a trump card for a semiconductor company

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With recession clouds beginning to withdraw, there seems to be a cheer in the semiconductor industry. What are the top trends one should watch out for in applications and devices powered by semiconductors this year?

Promising trends in the application area would include energy-efficient applications, which are very attractive and relevant considering the current circumstances (the ongoing climate debate, Obama's recent address stressing on clean energy, etc). We also see a huge trend favouring security related applications like payments, automatic fare collection and electronic identification. There is an increasing demand for safety in any kind of transaction. For instance, there is a trend right now in the PCB domain to identify the manufacturer and use technology to authenticate that company as the manufacturer of the board. Be it at an individual level, component level or PCB level, authentication solutions will be in demand.

On a macro level, we see the economies in Asia growing faster than the western world, making the Asian market crucial for any semiconductor player. Another macro trend that we see is the increase in optimised solutions. Today, it is no longer a generic component or process that can help you fix all the problems. Rather, a dedicated solution for a particular application area is making the world more sophisticated.

In devices, we predict a movement towards building more functionality on top of the standard complementary metal-oxide semiconductor (CMOS). Moore's law was about pushing the envelope for plain vanilla CMOS, wherein we went from 90 nanometres (nm) to 65 nm to 45 nm and now

32 nm, and that may continue. However, the trend will be more towards adding functionalities to the already present technology because if you are to create an optimised solution (a macro trend in application domain), you need to interface to it, and that most often is not CMOS voltage but a high voltage in a very robust and harsh environment. Take, for instance, a charger that you want to connect to a battery. It needs a technology allowing 40V, 60V or even higher voltages. Similarly, the drivers in solar streetlighting need up to 600 volts and also some control logic. Such complexities need special technologies that enable the outside world, which could be RF, high-voltage temperature sensors or pressure sensors, to connect to the inside world of digital computing. We call the technology that combines signal-processing capabilities with a number of the interfaces as 'HPMS' and predict this to be the next trend as it is essential to get an optimised solution for all types of interfaces.

In India, what are the key areas that will drive the semiconductor market?

We see tremendous potential in e-identification market in India. We see a good market for low-cost identification solutions, near-field communication (NFC) and services, radio frequency identification (RFID), smartcard products, and infrastructure components such as ICs and modules for terminals. Another area having great potential is HPMS-based solutions.

However, the Indian electronics ecosystem is still in a nascent stage. How can the Indian semiconductor eco-